RUDRA CHAVDA

rudrabchavda@gmail.com | LinkedIn: rudra-chavda | Portfolio: rudra.chavda.online | U.S. Permanent Resident

EDUCATION

University of California, Santa Barbara — Expected Spring 2028
Bachelor of Science, Computer Engineering — **GPA: 3.86**

University of California, Santa Cruz — 2024 – 2026
Bachelor of Science, Computer Engineering — **Honors:** 6x Dean's List

Coursework: Systems programming, computer architecture, operating systems, data structures and algorithms, and software engineering.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, TypeScript, JavaScript, SQL, RV Assembly
Frameworks	Next.js, React, Vue.js, Spring Boot, Node.js, Py4Web, Tailwind CSS, Angular
Tools	Git, Docker, Kubernetes, MySQL, Flyway, Microsoft Graph API, JUnit, CARLA, GDAL
Concepts	LLM Agents, Function Calling, OAuth 2.0, Reinforcement Learning, System Design, Data Structures

RELEVANT EXPERIENCE

Ascend Cargo — Jun 2026 – Sep 2026
AI/ML Software Engineering Intern — *Remote, San Francisco Bay Area*

- Architected an enterprise **LLM agent framework** utilizing **Java 23, Spring Boot 3.4, and OpenAI function calling**, featuring annotation-driven tool registration and dynamic JSON-Schema generation for targeted intent routing.
- Engineered a natural-language shipment management system, diagnosing and resolving a synchronous automation bottleneck that reduced database execution latency from **12 minutes to under 30 seconds**.
- Designed a Role-Based Access Control (RBAC) authorization pipeline enforcing strict per-user AI limits (**NONE/READ/WRITE**) and integrated a **Microsoft Graph API** inbound email channel via **OAuth 2.0** for authenticated, remote **shipment creation and updates**.
- Hardened agent fault tolerance by implementing explicit refuse-and-instruct error contracts, and authored **122 JUnit/Mockito unit tests** alongside a custom CLI harness to validate state mutations against the database.

AIEA Lab (UCSC) — Dec 2025 – Apr 2026
Undergraduate Researcher — *Onsite, Santa Cruz*

- Benchmarked and wrote Reinforcement Learning (RL) algorithms including **DQN, DDQN, A3C, and PPO** to evaluate autonomous vehicle safety parameters within the Robustifying AVs project.
- Deployed and scaled the **CARLA** autonomous driving simulator in a headless environment across the **Nautilus Kubernetes** hyper-cluster, isolating edge-case failure modes via post-training reward function visualization.

CA-Futures (UCSC EEB & Baskin Engineering) — May 2025 – Sep 2025
Software Developer — *Hybrid, Santa Cruz*

- Engineered a full-stack climate modeling platform utilizing **Vue.js** and **Py4Web**, delivering interactive web maps that enable researchers to visualize and compare predictive ecological shifts across multiple long-term climate scenarios.
- Architected an asynchronous geospatial pipeline utilizing **GDAL, Rasterio, and Geopandas** to intercept user-drawn map polygons, compute dynamic habitat connectivity, and implement output caching to eliminate redundant backend execution.

Opennote (YC S25) — Jul 2025 – Feb 2026
Campus Growth Lead — *Onsite, Santa Cruz*

- Spearheaded campus integration strategies, driving a **30% increase** in user engagement and securing **50+ daily active users** by organizing targeted technical workshops on TypeScript user-agent API flows.
- Co-directed the UCSC Cruzhacks integration, managing vendor partnerships and allocating **$1,500+** in technical resources to drive platform adoption among computer science students.

PROJECTS

Better Notes | *Next.js 15, TypeScript, Convex, Clerk, Tailwind* Spring 2025

- Architected a full-stack, real-time note-taking application, integrating **Clerk** for secure user authentication and **Convex** for reactive data synchronization and backend storage.
- Implemented a rich-text editing experience utilizing **BlockNote and ProseMirror**, enabling seamless markdown rendering and dynamic block-based study workflows for end-users.
- Scaled application to **100+ active users** within 3 months of Vercel deployment, optimizing database queries to maintain **sub-50ms read/write latency**.

DivExplorer Bias Detection | *Python, Pandas, ML, Shapley Values* Fall 2025

- Engineered an automated data analysis pipeline utilizing **FP-growth and Apriori** frequent pattern mining algorithms in Pandas to extract and evaluate statistically diverging subgroups within large datasets.
- Implemented statistical redundancy pruning thresholds and computed **Shapley values** to quantify specific feature contributions, isolating anomalies and reducing high-overlap duplicate discoveries.
- Quantified machine learning classification bias by measuring divergence metrics (e.g., **false-positive disparities**) across demographic intersections, establishing rigorous baselines for model fairness.

Personal Portfolio | *Next.js 16, TypeScript, Three.js, Motion, Tailwind* Winter 2024

- Developed a high-performance developer portfolio combining Next.js server-side rendering with complex 3D WebGL scenes using **Three.js** and declarative animations via **Motion**.
- Engineered a custom technical content rendering pipeline, integrating **KaTeX** for complex mathematical notation and dynamic syntax highlighting for interactive code blocks.
- Built a responsive, accessible component architecture utilizing **Tailwind CSS v4 and Shadcn UI** primitives, optimizing component hydration to maintain a consistent **60 FPS** experience.